

Use gradient-free when:

- Non-differentiable: gradient does not exist
- Black-box f: no closed-form expression
- Noisy f: gradient estimate corrupted by noise

$$x_{k+1} = \bar{x} + 2(\bar{x} - x_k)$$

Stop when:

- Simplex size: $\max\{x_i - x_j\} < \epsilon$
- Fn. value spread: $\max\{f(x_i)\} - f(x_j) < \epsilon$
- Iteration limit: $k > k_{\max}$ or $N_{\text{eval}} > N_{\max}$

$x_g = \frac{x_a + x_b + x_c}{3}$

$x_d = x_g + 2(x_g - x_c)$

$x' = x_d + \frac{1}{2}(x_c - x_d)$

- Reflection x_d
- Expansion x_d
- Contraction x_d
- Shrink x_k

$$x_{k+1}^{(i)} = x_k^{(i)} + v_{k+1}^{(i)} \Delta t, \quad v_{k+1}^{(i)} = \alpha v_k^{(i)} + \beta \frac{x_{\text{best}}^{(i)} - x_k^{(i)}}{\Delta t} + \gamma \frac{x_{\text{best}} - x_k^{(i)}}{\Delta t}.$$

Solution:

- $\mathbf{x}_i^{(k)}$: position of particle i at iteration k
- $\mathbf{v}_i^{(k)}$: velocity of particle i at iteration k
- $\mathbf{x}_i^{\text{best}}$: best position ever visited by particle i (personal best)
- $\mathbf{x}_{\text{best}}$: best position ever found by any particle in the swarm (global best)
- α : inertia weight (momentum of previous velocity)
- β : cognitive coefficient (attraction to personal best)
- γ : social coefficient (attraction to global best)
- Δt : time step; k = iteration index; i = particle index

(Evaluate fitness K_i)
 For each individual,
 select parents with
 probability proportional
 to fitness
 (reproduce what
 is successful)

Put selected parents
Choose a random cut
point and swing tails
to produce offspring.
(Wear uniform or
bed-gown clothes.)

with small probability $p_0 \ll 1$, randomly change one gene. Power-law persistence convergence and maintain genetic diversity.

1999

Dis: noisy f evals; no convergence guarantee. Tuning required (β_1, β_2 , pop size)

col 3 placeholder

col 4 placeholder